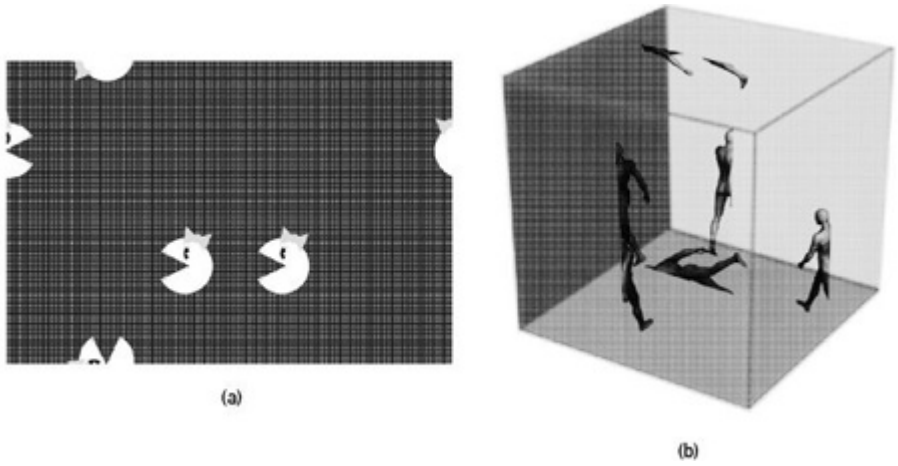


Figure 8.5 **(**a**)** A video game screen is flat (in the sense of "uncurved") and has a finite size, but contains no edges or boundaries since it "wraps around." Mathematically, such a shape is called a *two-dimensional torus.* **(**b**)** A three-dimensional version of the same shape, called a *three-dimensional torus,* is also flat (in the sense of uncurved) and has a finite volume, and also has no edges or boundaries, because it wraps around. If you pass through one face, you enter the opposite face.

To picture it, think of one of those video games in which the screen appears to have edges but in reality doesn't, since you can't actually fall off: if you move off the right edge, you reappear on the left; if you move off the top edge, you reappear on the bottom. The screen "wraps around," identifying top with bottom and left with right, and in that way the shape is flat (uncurved) and has *finite* size, but has no edges. Mathematically, this shape is called a *two-dimensional torus;* it is illustrated in Figure 8.5a.[12] The three-dimensional version of this shape —a three-dimensional torus—provides another possible shape for the fabric of space. You can think of this shape as an enormous cube that wraps around along all three axes: when you walk through the top you reappear at the bottom, when you walk through the back, you reappear at the front, when you walk through the left side, you reappear at the right, as in Figure 8.5b. Such a shape is flat—again, in the sense of being uncurved, not in the sense of being like a pancake—three-dimensional, finite in all directions, and yet has no edges or boundaries.

Beyond these possibilities, there is still another shape consistent with the symmetric expanding space explanation for Hubble's discovery. Although it's hard to picture in three dimensions, as with the spherical